

1. Administrative & Study Identification

Full Study Title: A Multicenter, Randomized, Double-blind, Parallel Group, Placebo-controlled Study to Evaluate the Efficacy and Safety of Iptacopan (LNP023) in Idiopathic Immune-complex-mediated Membranoproliferative Glomerulonephritis (IC-MPGN) **Short Title/Acronym:** APPARENT **Protocol Number:** APPARENT **Phase 3 NCT Number:** NCT05755386 **Trial Status:** Recruiting (Currently recruiting participants) **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Iptacopan hydrochloride (LNP023) **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of iptacopan versus placebo on proteinuria reduction measured as 24-hour urine protein-to-creatinine ratio (UPCR) at 6 months, and to evaluate the effect of iptacopan on proteinuria at 18 months. **Secondary Objectives:** To evaluate the change from baseline in estimated glomerular filtration rate (eGFR), assess the change in eGFR from the 12-month visit to the 18-month visit in the placebo arm, evaluate participants achieving a composite renal endpoint ($\leq 15\%$ reduction in eGFR + $\geq 50\%$ reduction in UPCR), and assess changes in patient-reported fatigue.

2.2 Background & Rationale Immune Complex-Mediated Membranoproliferative Glomerulonephritis (IC-MPGN) is a rare kidney disorder (Semantic Type: Disease or Syndrome) characterized by immune complex deposition within the glomeruli, the kidney's filtering units. These deposits, composed of antibodies and antigens, trigger inflammation and cellular proliferation, leading to a distinctive "tram-track" or "double contour" appearance of the glomerular basement membrane on kidney biopsy. This damage impairs the kidneys' ability to filter waste products from the blood, often resulting in proteinuria, hematuria, and progressive decline in kidney function that can advance to end-stage renal disease. Alternative complement pathway (AP) dysregulation is believed to underlie the clinical manifestations and progression of IC-MPGN. Currently, there are no approved targeted treatments. Iptacopan hydrochloride is a highly potent, oral proximal complement inhibitor that specifically binds to factor B, effectively inhibiting the AP to halt disease progression and reduce proteinuria. Upon completion of study treatment, participants will have the option to discontinue iptacopan and enter a 30-day safety follow-up or

transition to an open-label extension study (CLNP023B12001B; NCT03955445).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (followed by an open-label period) **Randomization:** 1:1

3.2 Planned Enrollment & Duration Target **NS:** Approximately 106 adult and adolescent participants (with a minimum of 10 adolescents) **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 02-Oct-2023 **Estimated Primary Completion:** 30-Jun-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 12 to ≤ 60 years at screening. 2. Biopsy-confirmed idiopathic IC-MPGN within 12 months prior to screening for adults, and within 3 years for adolescents. 3. Proteinuria measured as UPCR ≥ 1.0 g/g and an estimated glomerular filtration rate (eGFR) ≥ 30 mL/min/1.73 m². 4. Stable, maximally recommended or tolerated dose of renin angiotensin system inhibitors (RASi) for at least 90 days. 5. Mandatory vaccination against *Neisseria meningitidis* and *Streptococcus pneumoniae*.

4.2 Exclusion Criteria 1. Secondary IC-MPGN (e.g., secondary to chronic infection, autoimmune disorders, or monoclonal gammopathies). 2. Any cell or solid organ transplant, including kidney transplantation. 3. Rapidly progressive crescentic glomerulonephritis or kidney biopsy displaying $>50\%$ interstitial fibrosis/tubular atrophy. 4. Body mass index (BMI) >38 kg/m² or body weight <35 kg at screening and randomization.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Iptacopan 200 mg twice daily (b.i.d.) (administered as 2×100 mg capsules for adolescents), or matching placebo. **Route of Administration:** Oral **Duration of Treatment:** 360 days (12 months), comprising an initial double-blind period followed by an open-label period (recently amended to allow up to 540 days (18 months) for certain cohorts depending on response criteria).

5.2 Concomitant & Prohibited Therapy Permitted: Maximally tolerated or recommended doses of renin-angiotensin system inhibitors (ACE inhibitors or ARBs) and mandatory vaccination against encapsulated bacteria. **Prohibited:** Other complement

pathway modifying agents, and initiation/intensification of unapproved corticosteroids (>7.5 mg/day) or immunosuppressants (except MPAs) within 90 days of study drug administration.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -90 to 0	Biopsy confirmation, baseline eGFR/UPCR, vaccination against encapsulated bacteria
Baseline	Day 1	Randomization, first dose, duplicate 24-hour urine collection
Interim	Day 90 & Day 180	24-h urine tests (UPCR), eGFR assessment, safety labs, transition to open-label (if applicable at Day 180)
End of Study	Day 360 / 540	Final efficacy endpoints (UPCR/eGFR), exit interview, option to transition to 30-day safety follow-up or open-label extension study (CLNP023B12001B; NCT03955445)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- Log-transformed ratio to baseline in UPCR (sampled from a 24-hour urine collection) at 6 months.
- Log-transformed ratio to baseline in UPCR at the 18-month visit (each study treatment arm).
- Log-transformed ratio to 12-month visit in UPCR at the 18-month visit in the placebo arm.

7.2 Secondary & Exploratory Endpoints

- * Change from baseline in eGFR.
- * Change in eGFR from the 12-month visit to the 18-month visit of the placebo arm.
- * Proportion of patients achieving a composite renal endpoint ($\leq 15\%$ reduction in eGFR + $\geq 50\%$ reduction in UPCR compared to baseline).
- * Change in patient-reported fatigue (FACIT-Fatigue).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring collected via patient reports, onsite visit evaluations, and protocol-defined laboratory testing. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours) as per GCP guidelines. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee (DMC) will periodically assess the safety of iptacopan throughout the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) principle for primary efficacy end points; participants discontinuing randomized medication continue to be followed. **Sample Size Justification:** A supplementary analysis will utilize a 'Bayesian dynamic

borrowing' approach, incorporating a robust mixture prior to distribution to allow dynamic borrowing of prior information from the APPEAR-C3G phase 3 study. **Handling of Missing Data:** Patients discontinuing the investigational product will contribute to the treatment effect evaluation under the intention-to-treat principle.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with International Council for Harmonization (ICH) E6 Guidelines for Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard onsite and remote monitoring executed by the sponsor. **Audit Trail:** Continuous independent safety evaluations by the DMC and centralized eCRF data locking protocols.

11. Study Identifiers & Governance

Posting ID: 127463961077050325 **Primary ID:** APPARENT **Registry Number:** NCT05755386 **Sponsor/Lead Organization:** Novartis **Study Title:** Study of Efficacy and Safety of Iptacopan in Participants With IC-MPGN **Official Regulatory Title:** A Multicenter, Randomized, Double-blind, Parallel Group, Placebo-controlled Study to Evaluate the Efficacy and Safety of Iptacopan (LNP023) in Idiopathic Immune-complex-mediated Membranoproliferative Glomerulonephritis (IC-MPGN)

12. Clinical Framework (Study Specific)

Primary Condition: Idiopathic Immune Complex-Mediated Membranoproliferative Glomerulonephritis (IC-MPGN) **Preferred Name:** Immune Complex Membranoproliferative Glomerulonephritis **Semantic Type:** Disease or Syndrome **MeSH Code:** D008580 **SNOMED ID:** 773950005 **Diagnosis Threshold:** Biopsy-confirmed IC-MPGN with UPCR ≥ 1.0 g/g and eGFR ≥ 30 mL/min/1.73 m² **Medical Methodology:** Proximal complement alternative pathway (AP) inhibition combined with maximally tolerated ACEi/ARB supportive care **Optimization Target:** Proteinuria reduction (24-h UPCR) and eGFR stabilization

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for IC-MPGN including maximal RAAS inhibition **Education Protocol (HCP):** Protocol-specific training including vaccination requirements against encapsulated bacteria **Education Protocol (Patient):** Onsite training focusing on 24-hour urine collection instructions

and strict dosing adherence **Quality Audit Mechanism:** Independent Data Monitoring Committee (DMC) safety assessments and intention-to-treat clinical tracking

14. Observation & Follow-Up Schedule

Total Duration: 360-day treatment period + 30-day safety follow-up (can be extended up to 540 days for the amended protocol Cohort 2) **Onsite Visit Cadence:** Day 1, Day 90, Day 180, and Day 360/540 **Remote Monitoring Frequency:** Periodic safety checks mapping to protocol standards **Checklist Review Interval:** Every scheduled primary visit (e.g., Q90 Days)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				